

Task 1: Deployment Model for Two Apps

1. Instagram — Public Cloud

Reason 1: Public cloud provides large-scale infrastructure that can support millions of users.

Reason 2: Resources can scale according to traffic and geographic demand.

2. Paytm — Hybrid Cloud

Reason 1: A hybrid approach can allow sensitive workloads and data to receive additional control while using public-cloud resources for scalable services.

Reason 2: Public cloud capacity can help handle large traffic spikes while more controlled infrastructure can be used for selected sensitive workloads.

Task 2: Deployment Model Comparison

Model	Pros	Cons
Public Cloud	Lower initial cost; highly scalable; easy deployment	Less direct infrastructure control; shared infrastructure; compliance considerations
Private Cloud	Greater control; customization; suitable for sensitive workloads	Higher cost; requires more management; scaling can be more difficult
Hybrid Cloud	Flexible; combines private and public resources; useful for handling traffic spikes	More complex management; integration challenges; potentially higher operational complexity

Community Cloud	Shared by organizations with common requirements; collaborative; can support specific compliance needs	Smaller scale; governance can be complex; limited compared with major public clouds
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Task 3: WhatsApp Cloud Infrastructure

WhatsApp is a globally distributed messaging service that requires high availability, scalability and reliable infrastructure. For an academic deployment-model classification, a public-cloud-oriented model is suitable because public cloud infrastructure can provide large-scale computing, storage and networking resources across geographic regions.

Public cloud infrastructure can help a service handle changing workloads and global user demand. It can also provide managed security, monitoring and infrastructure services while reducing the need to maintain all physical infrastructure directly.

Note: Real-world WhatsApp infrastructure is much more complex than simply labeling it as one deployment model, and its infrastructure has evolved over time.

Task 4: IPL Ticket Booking Platform

Recommended Model: Public Cloud

A public cloud would be suitable because an IPL ticket-booking platform can experience extremely high traffic when tickets become available.

Risks

1. Traffic spikes: Extremely high traffic could cause service degradation.
2. Security attacks: Attackers could target the platform with DDoS attacks or account attacks.

Mitigation

Use auto-scaling, load balancing, CDN/WAF protection, rate limiting and monitoring to handle traffic and protect the application.